

Preliminary Analysis

To assess for possible differences between the conditions at baseline, demographic characteristics (age, gender, sexual orientation, race, year in school, Hispanic ethnicity), and baseline symptom measures (BDI-FS, PHQ-9, DASS-SF, PANAS, BADS-SF) will be compared. Chi-square analysis will be used for categorical measures, and independent samples *t*-tests will be used for continuous measures. Finally, the number of missing daily survey responses would be compared across conditions using an independent samples *t*-test, and the proportion of missing mid-treatment, post-treatment, and follow-up survey responses would be compared across conditions using Chi-square tests.

Outcome Analysis

Linear mixed effects models will be used to model change in symptom measures (BDI-FS, PHQ-9, DASS-SF, PANAS, BADS-SF) from baseline to each of the mid-treatment, post-treatment, and follow-up timepoints. Models will include random effects for the intercept and fixed effects for the intercept, condition (BA vs. SM), three time contrasts (baseline vs. each subsequent timepoint), and interactions between condition and each of the time contrasts. This model will allow for outcome variable scores to be estimated at each individual time point for each group. Terms of primary interest include the main effect of treatment, which indicates the difference in symptoms across conditions on average across all time points, as well as the interaction between time and condition, which indicates the difference in symptom change across conditions. The main effect of treatment in these models will be interpreted to indicate the overall effect of the intervention on the outcomes, as recommended by Mallinckrodt et al. (2008). Additionally, the omnibus treatment-by-time interaction will be interpreted to indicate the overall difference across conditions in symptom change from baseline. Additionally, planned analyses include examining the time-point specific interactions between treatment condition and time (i.e., interaction of treatment condition and each of baseline-to-mid-treatment time, baseline-to-post-treatment time, and baseline-to-follow-up time) and examining the simple effect of treatment at each time-point after baseline (mid-treatment, post-treatment, follow-up). These analyses will allow indexing the overall treatment effect as well as the effect of treatment on symptom change from baseline to specific time-points. Missing data in all models will be handled using full information maximum likelihood estimation.

Exploratory Analysis

To examine treatment satisfaction, I will calculate descriptive statistics on the Mental Health Satisfaction Questionnaire scores within each condition and compare means using an independent samples *t*-test. Additionally, to examine daily changes in PANAS and BADS-SF scores during the 14-day intervention period, I will plot average scores over time within conditions and conduct mixed effects models with linear and/or nonlinear time effects to capture change over time in each condition. Given the richness of this longitudinal daily data, including the possibility for a range of nonlinear change patterns, the modeling approach will be chosen following visualization of scores over time, time series residual diagnostics (e.g., tests of residual autocorrelation), model fit criteria (e.g., chi-square difference tests, Akaike Information Criteria, and Bayesian Information Criteria comparisons across candidate models). Candidate models

include linear mixed models with and without polynomial growth terms, generalized additive models, and linear mixed models with autoregressive and/or moving average error structures. The key goal of these analyses will be to characterize as accurately and parsimoniously as possible the change in daily symptoms over time within the two conditions. Missing data in all models will be handled using full information maximum likelihood estimation.

Finally, to examine mechanisms of change, I will conduct structural equation models examining changes in BADS-SF scores from pre-treatment to mid-treatment as a mediator of the effect of treatment condition on change in BDI-FS scores from pre-treatment to post-treatment. As with the daily diary data, this longitudinal analysis could be modeled in a number of possible ways; with this in mind, a mediation model will be selected based on examining model fit criteria and convergence for several possible models. One relatively simple possible model would involve specifying treatment condition as a predictor of mid-treatment BADS-SF scores, and modeling both treatment condition and mid-treatment BADS-SF scores as predictors of post-treatment BDI-FS scores, and examining the indirect effect using bootstrapping. On the other hand, a more complex model would modeling longitudinal state-like relationships between BADS-SF and BDI-FS over the three time-points using either random intercepts cross-lagged panel modeling or latent change score modeling, and examining if the longitudinal relationship between state-like changes in BADS-SF explain treatment condition's effect on BDI-FS. Again, the goal would be to select the most accurate and parsimonious model that will fit the data. Missing data in all models will be handled using full information maximum likelihood estimation.